

WHITEMORE NGWIRA

PRINCIPAL TECHNOLOGY ARCHITECT & AI SYSTEMS ENGINEER

Cloud Architecture (AWS Primary) | AI Systems & Gateways | Networks, Databases & Security | DevOps & IaC

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PROFESSIONAL SUMMARY

Principal Technology Architect & AI Systems Engineer with 15+ years of experience spanning cloud architecture (Amazon Web Services primary, with Microsoft Azure and Google Cloud Platform), systems architecture, advanced networking, multi-engine databases, zero-trust security, AI systems & AI gateways (LiteLLM, Cloudflare AI Gateway), and automated operations. I work across discovery, architecture, engineering, and prototype-to-production delivery, connecting business requirements with scalable infrastructure, APIs, automation, and live systems. A central foundation of my practice is treating broadcast production, multimedia pipelines, and industrial operations as high-stakes systems engineering: designing and automating the technical lifecycle from sensor/capture ingest and encrypted cloud storage (Amazon S3, AWS KMS) through distributed compute pipelines, automated quality control (QC), compliance delivery, and archival preservation (S3 Glacier). Cross-sector track record across insurance (EarCodeX), industrial mining telemetry (Socinga Smart Mining Platform), education platforms (Cinetrans, Oasis College), e-commerce, property/construction, and 21 major broadcast/film productions. Proven expertise in modular Terraform IaC blueprints, AWS SAM, Serverless Framework, Kubernetes (EKS v1.32), CI/CD, least-privilege security (AWS IAM/RBAC/OIDC), CloudWatch/CloudTrail observability, and governed AI-assisted delivery to release capacity, expand revenue, and protect profitability.

CORE EXPERTISE

- **Cloud Architecture & Infrastructure (AWS Primary):** Amazon Web Services (AWS); Compute (EC2, Lambda, ECS Fargate, EKS v1.32, Auto Scaling); Multi-Tier Storage (Amazon S3 Intelligent-Tiering, Versioning & Lifecycle, EBS gp3, EFS, FSx for Lustre, S3 Glacier); Disaster Recovery & Cross-Region Replication; Microsoft Azure & GCP; AWS Budgets & Cost Explorer.
- **Networking & Distributed Connectivity:** Amazon VPC (Multi-AZ Subnets, Route Tables, NAT Gateways, Internet Gateways); AWS Transit Gateway (Multi-VPC Hub & Hybrid Architecture); VPC Endpoints (S3 Gateway, KMS/CloudWatch/STS Interfaces); Amazon Route53 (DNS, Health Checks, Latency Routing); Amazon CloudFront CDN; AWS Global Accelerator; Tailscale Distributed Mesh VPN (Zero-Trust WireGuard Network).
- **Security, Governance & Compliance:** Defence in Depth; AWS IAM (Least-Privilege Policies, Roles, MFA, OIDC CI/CD Federation); AWS KMS (Envelope Encryption); AWS WAF (Web ACLs, SQLi, XSS, Rate Limiting, Geo-Blocking); AWS Shield (DDoS); AWS GuardDuty (ML Threat Detection); AWS Inspector; AWS CloudTrail (Multi-Region Immutable Audit Logging); AWS Macie; AWS Cognito; S3 Object Lock.
- **Databases, Data Lakes & Analytics:** Relational & NoSQL Systems; Amazon DynamoDB (Key-Value & Document Storage); AWS RDS (PostgreSQL Multi-AZ, Read Replicas); ElastiCache Redis (In-Memory Caching); DocumentDB; Amazon Neptune (Graph Database); Amazon Timestream (Telemetry); AWS Redshift; AWS Lake Formation, AWS Glue & Amazon Athena (Serverless SQL Analytics).
- **AI Engineering, RAG & AI Gateways:** Generative AI & LLMs; AI Gateways (LiteLLM for unified multi-model routing, load balancing, token cost tracking, provider failover; Cloudflare AI Gateway for edge caching across 300+ cities, rate limiting & request logging); Amazon Bedrock (Knowledge Bases, Intelligent Agents, Guardrails); Vector RAG (Titan & Cohere Embeddings, OpenSearch Serverless); Agentic Workflows; Responsible AI Governance.
- **DevOps, IaC & Delivery Automation:** Terraform (37 Modular Blueprints, S3 Remote State, DynamoDB Lock Table, Drift Detection); AWS SAM & Serverless Framework; Kubernetes (EKS v1.32, Pod Disruption Budgets, HPA, IRSA); Helm Charts (12 Production Templates); Docker (Multi-Stage Builds, ECR); GitHub Actions & GitLab CI/CD (6-Stage Dual Pipelines, OIDC); Bruno & OpenAPI Specifications (53 Test Suites); CloudWatch Alarms to Slack/Teams via SNS.
- **Systems & High-Throughput Media Architecture:** Enterprise & Solutions Architecture; ADRs; NFRs; Telemetry & IoT Ingest; Secure Media Ingest & Object Storage; Automated Processing Pipelines; Technical Quality Control (QC); Broadcast Playout Standards (DStv, Netflix, SABC); DaVinci Resolve Studio; Avid Media Composer; Adobe Creative Cloud; Digital Intermediate (DI) Infrastructure; Deep Archive Lifecycle Management.

PROFESSIONAL EXPERIENCE

PRINCIPAL SYSTEMS ARCHITECT & DIGITAL INFRASTRUCTURE LEAD

Socinga Africa | Johannesburg, South Africa | January 2025 – Present

Lead enterprise architecture, cloud infrastructure, governed AI automation, and cross-sector digital delivery across insurance, media, mining, and foundation divisions.

- **Enterprise Platform Architecture:** Architected Socinga Africa's cloud-first enterprise platform across four operating divisions. Translated operational requirements into as-is and target-state architectures, domain, API and data boundaries, data models, RBAC, and security by design, performance, scalability, and resilience; recorded ADRs and deployed version-controlled Next.js, React, and TypeScript applications, eliminating cross-border processing delays and handovers.
- **Governed AI Document Automation:** Designed and operationalised an enterprise LLM-assisted document-intake workflow from prototype into live operation under Responsible AI governance: ingested form, email, and CRM data; implemented RAG, extraction, classification, structured outputs, grounding, and accuracy evaluation; routed exceptions and approvals to accountable staff; maintained audit logs, observability, and rollback, ensuring authorised staff retain final responsibility for regulated decisions.
- **AI-Accelerated SDLC & Quality Gates:** Introduced specification-led, AI-accelerated engineering across the complete software development lifecycle (SDLC)—spanning requirements, architecture records, prototyping, coding, refactoring, automated testing, debugging, documentation, and release QA; enforced mandatory human code review, Git workflows, security checks, CI/CD pipelines, and deployment gates, rejecting unverified outputs before release.
- **Commercial Operations & Capacity:** Automated lead follow-up, onboarding, and digital media production workflows from brief intake through scheduling, metadata tagging, asset versioning, and first-pass QA; reduced administration and rework, released billable capacity, expanded revenue capacity, and protected margins, directly supporting profitability across an active portfolio valued internally above ZAR 5 million (tracking 126+ partnerships and 18+ initiatives).

INDEPENDENT SYSTEMS & CLOUD ARCHITECT | PRODUCTION TECHNOLOGY CONSULTANT

N.White Systems (Independent Practice) | South Africa & SADC Region | August 2009 – Present

Concurrent consultancy delivering remote and asynchronous cross-border cloud architecture, systems engineering, industrial telemetry, media production technology, and technology advisory.

- **Client Discovery & Solutions Strategy:** Led client discovery, technical demonstrations, AI use-case workshops, and architectural estimations; delivered production blueprints, decision records (ADRs), risk/ROI frameworks, and execution roadmaps that accelerated stakeholder alignment and de-risked complex cross-border delivery.
- **Full-Stack SaaS & Cloud Platforms (AWS/Azure/GCP):** Architected and delivered full-stack platforms using Next.js, React, TypeScript, Node.js, Python, Tailwind, Supabase/PostgreSQL, and REST APIs; deployed across AWS (primary), Azure, GCP, Cloudflare, and Vercel with automated CI/CD and WCAG accessibility, achieving resilient high-availability performance.
- **AI Gateways & Agentic Automation Workflows:** Architected and integrated AI gateways with LiteLLM (multi-model routing, token cost tracking, Bedrock/Anthropic/OpenAI failover) and Cloudflare AI Gateway (edge caching across 300+ cities, rate limiting); operationalised human-supervised agentic workflows across forms, email, and CRM to reduce response times and protect margins.
- **Insurance Platform Architecture (EarCodeX):** Delivered EarCodeX from prototype to production as an AWS cloud-native InsurTech platform; engineered claims administration, automated document intelligence, reconciliation services, immutable audit trails, and auditable human review, establishing robust privacy controls for regulated data.

N.WHITE SYSTEMS (INDEPENDENT PRACTICE) — CONTINUED

- **Infrastructure as Code, Security & Networking Blueprint:** Engineered multi-tier AWS environments using Terraform (37 modular blueprints, S3 remote state, DynamoDB locking) and SAM; implemented Transit Gateway hybrid connectivity, Route53 DNS, S3 lifecycle policies, KMS envelope encryption, IAM least-privilege, and Tailscale zero-trust VPN, eliminating configuration drift and securing distributed access.
- **Multi-Engine Databases & Data Lakes:** Architected multi-engine data tiers across Amazon DynamoDB, AWS RDS PostgreSQL (Multi-AZ), ElastiCache Redis, DocumentDB, Neptune, and Timestream; built serverless data lake pipelines via AWS Lake Formation, Glue, and Athena, enabling sub-second analytical queries across heterogeneous operational stores.
- **Mining Telemetry & Operational Data Flows (Socinga Smart Mining):** Designed shaft-to-mill industrial telemetry and IoT sensor data architectures from extraction points to processing mills, engineering the technical foundation for the [Socinga Smart Mining Platform](#) to improve real-time operational visibility and executive decision-making.
- **Education Technology Platforms (Cineterms & Oasis College):** Built and deployed live edtech platforms (Cineterms and Oasis College) from prototype to production using Next.js, TypeScript, Claude API, Supabase, and Tailwind; implemented multi-agent orchestration and learner workflows with human oversight to guarantee reliable, controlled educational delivery.
- **End-to-End High-Throughput Media Systems (21 Broadcast Productions & SAMF):** Architected high-throughput camera-to-cloud media pipelines, checksum-verified ingest, distributed post-production compute, and S3 Glacier archival preservation across 21 major productions (Netflix, MultiChoice Studios, SABC, 2010 FIFA World Cup), achieving automated quality control (QC) and playout compliance.
- **Digital Intermediate (DI) & Broadcast Playout Enablement (Our Perfect Wedding Zimbabwe):** Engineered the technical Digital Intermediate (DI) colour grading pipeline and master playout integration for Our Perfect Wedding Zimbabwe on DSTv Zambezi Magic (Channel 162); conducted technical post-production workshops for Ngoda TV broadcast engineers to ensure strict network compliance.
- **Repeatable Digital Content Workflows (Brand South Africa):** Engineered repeatable end-to-end digital content production and automated metadata tagging workflows for Brand South Africa's 20th anniversary video series (Ebony+Ivory), streamlining multi-version asset preparation and post-production delivery to meet compressed campaign milestones.
- **Broadcast Systems Modernisation & Playout Architecture:** Advised SADC regional broadcasters on studio modernisation and engineered automated broadcast playout and on-air identity systems across six television networks (Soico Televisão, TV Miramar, Zodiak TV, Times TV, Supa TV, and JIT TV), ensuring 100% transmission stability and regulatory compliance.

DIRECTOR OF DIGITAL ARCHITECTURE & TECHNOLOGY

Reason GTMC | Johannesburg, South Africa | January 2023 – January 2025

Led enterprise platform architecture, digital product engineering, and high-throughput multimedia delivery systems for enterprise clients.

- **Enterprise Web Architecture & Design Systems:** Architected Next.js and React enterprise applications, robust API gateway integrations, and component design systems for financial services, property, and public sector clients, embedding WCAG accessibility, Core Web Vitals optimisation, and technical SEO.
- **Governed AI Delivery & Workflow Automation:** Deployed human-supervised generative AI pipelines for rapid requirements decomposition, automated metadata tagging, and first-pass release QA, shortening iteration cycles while maintaining strict governance and factual verification before production deployment.
- **Multidisciplinary Systems Engineering Leadership:** Directed cross-functional engineering and technical teams through Agile delivery cycles, managing technical specifications, cloud resource planning, broadcast standards compliance, security gates, and project margins.

LEADERSHIP, GOVERNANCE & INDUSTRY ROLES

- Former Chairperson: SAFAITS; Former Chairperson & Convenor: SAFREA; Former Co-Chairperson: SASFED; Former Co-Chairperson & Jury Member: SAFTAS (2021–2024).
- Former Board Member & Secretary: SAAVC; Former Executive Committee Member: SAGE; Jury Member: Academy of Motion Picture Arts and Sciences (AMPAS), 94th Awards Cycle (2022).

EDUCATION & TECHNICAL CREDENTIALS

Academic Qualifications

Executive Master of Business Administration (EMBA) - In Progress. | Bachelor of Arts Honours - Journalism & Media Studies. | Diplomas in Business Administration, Marketing, and Information Technology.

Technical Certifications

Generative AI Fundamentals - Google | Applied AI for Digital Transformation - IBM | Prompt Engineering - DeepLearning.AI | AI for Business Leaders - Udacity | Google Analytics IQ - Google | Content Marketing Certification - HubSpot | Full-Stack Development Certification - Udemy.

VERIFIED ONLINE EVIDENCE & CODE REPOSITORIES

- **N.White Systems Platform:** <https://nwhite.systems/> | **Recruiter Portfolio Review:** <https://nwhite.systems/portfolio-review>
- **GitHub Repositories:** <https://github.com/whitemorengwira> | **Stack Overflow Profile:** <https://stackoverflow.com/users/32809886/whitemore-ngwira>
- **Hugging Face Hub:** <https://huggingface.co/nwhite-systems> | **DEV Community:** https://dev.to/nwhite_systems
- **Medium Publications:** https://medium.com/@nwhite_systems | **LinkedIn Profile:** <https://linkedin.com/in/whitemorengwira>
- **Upwork Verified Profile:** <https://upwork.com/freelancers/~01a67ba5995f66ae84> | **Dribbble Design Systems:** <https://dribbble.com/whitemore-ngwira>

SELECTED PRODUCTION EVIDENCE & PORTFOLIO ARTIFACTS

- **EarCodeX – Live InsurTech Cloud Platform:** Production policy/claims administration, automated reconciliation, and auditable human approvals: <https://earcodex.vercel.app/>
- **N.White Systems Platform:** Architecture, cloud engineering, full-stack software, and governed AI ecosystem: <https://nwhite.systems/>
- **Socinga Smart Mining – Industrial Telemetry:** Shaft-to-mill capture points, sensor telemetry, and operational data architecture: <https://www.socinga.africa/mining/smart-mining>
- **High-Throughput Media Systems:** Camera-to-cloud pipelines, automated QC, and broadcast playout mastering across 21 productions for Netflix, MultiChoice Studios, and SABC: <https://nwhite.systems/my-portfolio/multimedia>
- **Cineterms & Oasis College – EdTech Platforms:** Production Claude API, Supabase, and Next.js platforms (<https://nwhite.systems/my-portfolio/cineterms> | <https://nwhite.systems/my-portfolio/oasis-college>).
- **SAFA ProMax – Public Source Platform:** Production Next.js/TypeScript platform with automated build verification and lint tests: <https://github.com/whitemorengwira/safa-promax>